

Final project – proposed topics

Choose one project from the following topics:

1. Stochastic computation using B-spline expansion:

This project focuses on representing the stochastic responses using B-spline basis function as alternative to generalised orthogonal polynomials and discuss the suitability of this method in dealing with Gaussian random inputs.

Ref: Christoph Eckert, Michael Beer, and Pol D Spanos. A polynomial chaos method for arbitrary random inputs using b-splines. Probabilistic Engineering Mechanics, 60:103051, 2020.

2. Multifidelity Monte Carlo:

The focus of this project is the implementation of a multifidelity Monte Carlo approximation to estimate the stochastic response of a system. Multi-level Monte Carlo exploits a hierarchy of coarse-grid approximations of the governing equations of the high fidelity model to construct variates. Multifidelity Monte Carlo on the other hand uses control variates to reduce the variance in Monte Carlo estimation, but relies on low-fidelity models for constructing these control variates. In particular, the aim of this project is to investigate the properties required for such low-fidelity models and the appropriate way to use the low-fidelity models constructed in concert to speedup stochastic computations.

Ref: Benjamin Peherstorfer, Philip S Beran, and Karen E Willcox. Multifidelity monte carlo estimation for large-scale uncertainty propagation. In 2018 AIAA Non-Deterministic Approaches Conference, page 1660, 2018.

Ref: A Aruna and Ranjan Ganguli. Multi-fidelity response surfaces for uncertainty quantification in beams using coarse and fine finite element discretizations. International Journal for Computational Methods in Engineering Science and Mechanics, pages 1–20, 2020.

3. Probability box approach for stochastic computations with unknown correlation structure:

This project focuses on the representation of epistemic uncertainty regarding the correlation structure of a random field. If the kernel function is unknown, its influence on the input field as well as on the quantity of interest (QoI) is important and should be studied in this project for different kernel functions (Ref 1). The necessity of acknowledging the lack of knowledge regarding the correlation function within a probability box (p-box) approach (Ref 2) should be discussed, e.g. by using an interval valued exponent ν in the Whittle-Matérn kernel.

Ref 1: K.-K. Phoon J. Ching. Impact of autocorrelation function model on the probability of failure. J. Eng. Mech., 145:04018123, 2019.

Ref 2: V. Kreinovich M. Beer, S. Ferson. Imprecise probabilities in engineering analyses. Mechanical Systems and Signal Processing, 37:4–29, 2013.

4. Stochastic analysis of 3D nonlinear stochastic problems:

This project considers a 3D problem (with a complex geometry) in the area of computational solid mechanics that is nonlinear and could be, for example, elastoplastic, hyperelastic, or a contact problem. You can choose a type of nonlinear problem, but the stochastic hyperelastic problem is recommended. The randomness is from Young's modulus and external force, where the Young's modulus is a Gaussian random field with given mean and covariance function, and the external force is related to a random variable. The goal is to, 1) simulate the random field with low- and high-dimensional random variables by means of the KL expansion, and input the simulated random field to commercial FEM software (any software you want or you can use your own codes), and 2) solve the stochastic nonlinear problem in the sense of MCS and output the results. Based on the above results, the PCE is used to approximate the obtained stochastic solutions with low-dimensional random inputs.

Some other suggestions of project topics:

5. Anisotropic correlation structure
6. Inverse problems, Bayesian approaches
7. Sparse grid approaches
8. Stochastic computations with POD
9. Non-Gaussian random field
10. Describing random loads, random load modeling
11. Stochastic homogenization
12. Fokker-Planck equation
13. Describing the lack of knowledge due to insufficient experimental data: stochastic approach vs. data-driven approach

Requirements:

1. You should finish one package. All codes should be kept and clear for verification.
2. Write one report. The report must be written in a formal way like a journal article with solid theory and clear implementation details.
3. In the end you should give a presentation of your work.